

Gold Volume-Surge Signal: Section C Interim Results

Velocity / Acceleration · Duration before Entry · Burst Phase

Source: final executed Section C in lup-2 · Close session (15:00-16:00 ET) · Control: baseline static GC_signal_30

1. Section C objective

Section C asks whether predictive information in gold-futures volume comes from the static surge level, from velocity / acceleration of that surge, from how long the surge has stayed elevated before entry, or from the burst phase (onset, rising flank, persistence, confirmed fade).

This is a one-axis test. The control is the original static GC total-volume baseline (30-bar prior mean, expanding normalization, close window, GLD horizons 1 / 5 / 10 / 30). Section A lookback winners, B uptick, MGC, and ETF constructions are not mixed into this screen.

Reading note. IC and conditional mean-forward screens are descriptive. Strategy statistics for the packaged C.2 survivor are provisional (non-overlapping long-only holds) and are not out-of-sample or production evidence.

2. C.1 Velocity / acceleration

Static measures how elevated the surge currently is. Velocity is the change in that level over 1 / 5 / 15 minutes (building vs cooling). Acceleration is the change in velocity (whether the build-up is strengthening or slowing). Lags require same-date, exact-minute continuity.

Pearson IC (close window)

Construction	@ 1m	@ 5m	@ 10m	@ 30m
Static	-0.0040	0.0010	0.0048	0.0166
Vel_1	-0.0145	-0.0066	0.0001	0.0008
Vel_5	-0.0034	-0.0010	-0.0000	-0.0019
Vel_15	-0.0084	-0.0028	-0.0007	0.0010
Acc_1	-0.0166	-0.0065	-0.0013	0.0001
Acc_5	-0.0039	-0.0031	-0.0025	-0.0049
Acc_15	-0.0031	-0.0005	-0.0000	0.0012

Spearman IC (close window)

Construction	@ 1m	@ 5m	@ 10m	@ 30m
Static	0.0069	0.0093	0.0043	-0.0009
Vel_1	0.0026	0.0055	0.0008	-0.0042
Vel_5	0.0057	0.0001	0.0043	-0.0085
Vel_15	0.0004	0.0049	0.0080	-0.0060
Acc_1	-0.0028	0.0012	-0.0062	-0.0050
Acc_5	-0.0007	-0.0065	0.0006	0.0039
Acc_15	0.0006	0.0017	0.0034	0.0005

C.1 Static vs Velocity / Acceleration (Close Window)

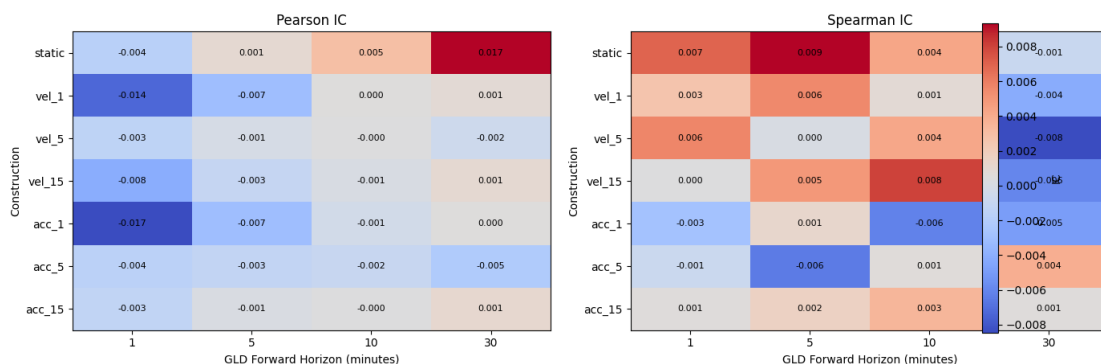


Figure: C.1 Pearson and Spearman IC heatmaps - static vs velocity / acceleration (close window).

Result: No dynamic construction beats static surge on the useful close horizons. At 30m, static Pearson IC is 0.0166 while velocity / acceleration Pearson ICs are near zero (about -0.005 to +0.001). Spearman results are similarly flat and do not favor vel/acc. Short-horizon Pearson ICs for vel_1 / acc_1 are slightly negative (cooling-consistent) but small.

Candidate decision: No new candidate from C.1. Velocity and acceleration do not improve on static surge under this screen.

3. C.2 Duration above threshold before entry

Once static surge first becomes unusually elevated, does waiting for it to remain elevated for N consecutive minutes before entering improve subsequent GLD returns?

Final entry-confirmation methodology

- Threshold (fixed, not optimized): lagged expanding historical 80th percentile of GC_signal_30.
- Burst: begins when surge crosses above threshold; ends when it falls below, the date changes, or minutes are not consecutive.
- Entry: one entry per burst at the first moment confirmation is reached - streak == 1, 2, 3, or 4 bars. This is waiting before entry, not trading every elevated minute.

Mean forward return (bps) at first confirmation

Confirm bars	@ 1m	@ 5m	@ 10m	@ 30m	n @ 30m
1 (onset)	0.09	-0.10	0.16	0.62	1,994
2	-0.06	0.06	0.37	1.16	647
3	0.13	0.17	0.25	1.23	277
4	0.48	1.85	1.52	3.94	129

Hit rates at 30m rise from 48.7% (1-bar) to 57.4% (4-bar). Entry counts fall sharply as confirmation lengthens - the sample / opportunity trade-off is material.

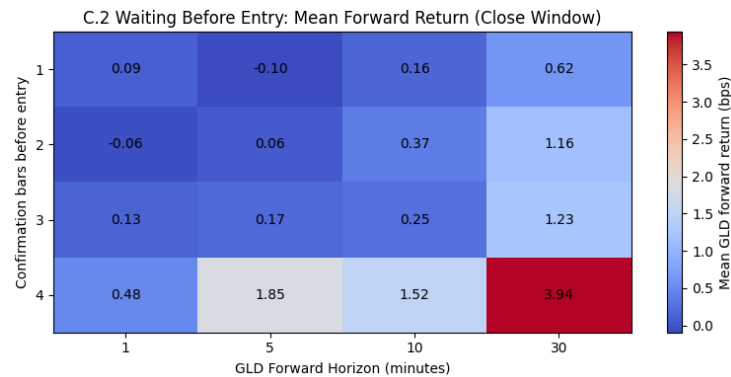


Figure: C.2 confirmation duration x horizon - mean GLD forward return (bps), close window.

Among the tested rules, 4-bar confirmation at the 30-minute GLD horizon is the strongest provisional screen result, but sparse. Only that rule is packaged; 1/2/3-bar rules are not retained.

Survivor package - C2_confirm4_h30 (provisional strategy diagnostics)

Avg bps/trade	Hit	Sharpe	Sortino	Trades	Days	Max DD (bps)
4.422007	58.26%	3.426639	9.110155	115	104	-100.672

Pearson / Spearman IC are not applicable to this categorical entry rule (reported as blank in the notebook). Metrics use non-overlapping 30m holds. Provisional and sparse - not a robustness or production claim.

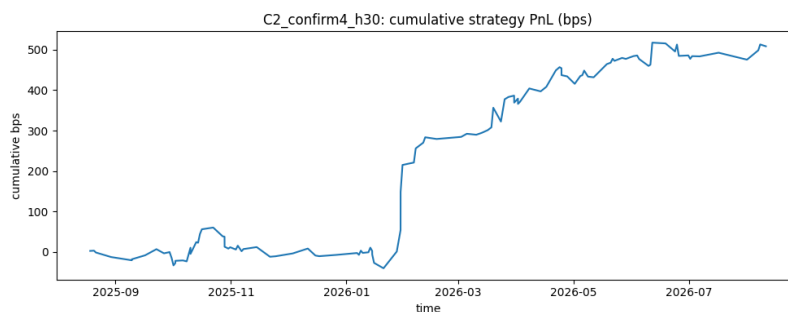


Figure: C2_confirm4_h30 provisional cumulative P&L (bps). Path is lumpy; gains concentrate in a few periods.

4. C.3 Burst phase

Conditional on an elevated burst, which state looks better for entry: onset, rising flank, ongoing persistence, or confirmed fade? C.2 asks how long to wait; C.3 asks what phase the burst is in.

- Onset: first above-threshold bar (streak = 1).
- Rising flank: elevated, streak ≥ 2 , and surge still rising vs the prior bar.
- Persistence: elevated with streak ≥ 2 (broad ongoing state; overlaps rising flank and some fade bars).
- Confirmed fade: still elevated; prior bar was rising; current surge has fallen from the prior bar.

Mean forward return (bps) by phase

Phase	@ 1m	@ 5m	@ 10m	@ 30m	n @ 30m
Onset	0.09	-0.10	0.16	0.62	1,994
Rising flank	0.01	0.17	0.46	1.76	569
Persist	0.19	0.41	0.70	1.68	1,173
Fade	0.22	0.32	0.57	0.86	501

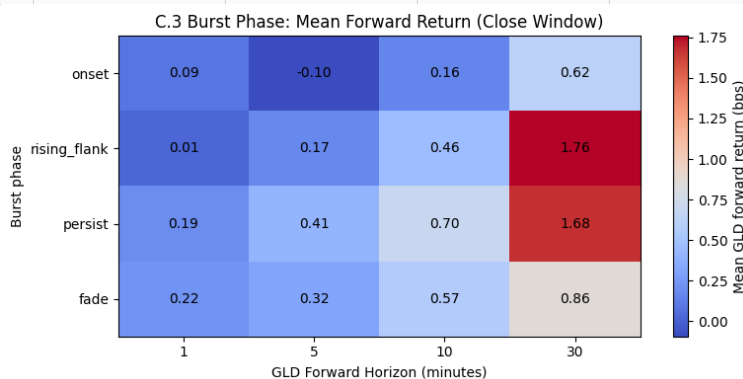


Figure: C.3 burst phase x horizon - mean GLD forward return (bps), close window.

Result: At longer horizons, rising flank (~1.76 bps @ 30m) and persistence (~1.68) exceed onset (~0.62) and confirmed fade (~0.86). Onset is the weakest longer-horizon state. This supports waiting past immediate onset (consistent with C.2) but does not introduce a separate packaged entry rule.

Candidate decision: No new candidate from C.3.

5. Section C decision

Question	Answer
Does velocity / acceleration improve static surge?	No. Static remains stronger on Pearson and Spearman IC.
Does waiting for persistence help?	Yes on the conditional screen: longer confirmation raises mean forward bps at 30m, with sharply fewer entries.
Which burst phase looks strongest?	Rising flank and persistence beat onset and confirmed fade at longer horizons.
Which C candidate(s) survive?	C2_confirm4_h30 only (provisional / sparse). C.1 and C.3 add no new packages.

Under this one-axis test, predictive content is better described as static elevation that has already been confirmed for a few minutes, not as velocity, acceleration, onset, or post-peak fade. The surviving confirmation filter will be evaluated later alongside other retained candidates; it should not be treated as standing alone on this screen.